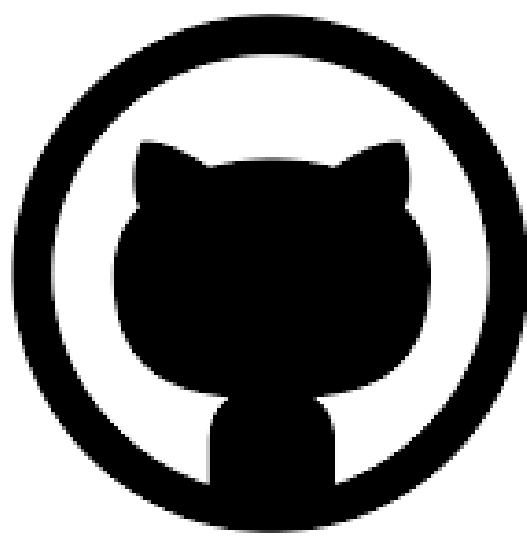




MINIGIT

Version Control System in C

★ DSA feels more meaningful when it builds something real.

★ MiniGit felt like real software, not just another lab task.
Commits, branches, diffs the things developers use every day.

★ Git is powerful—and mysterious—so we wanted to understand it from the inside.

★ One project, all major data structures.
Linked lists, hash tables, stacks, queues, trees, graphs, even LCS (DP).

★ A unique, impressive, resume-worthy project.

There's something exciting about saying, "We built our own Git"

By MIS
612415123
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PROBLEM STATEMENT

Git Learning Challenges for Students

Synchronization Issues

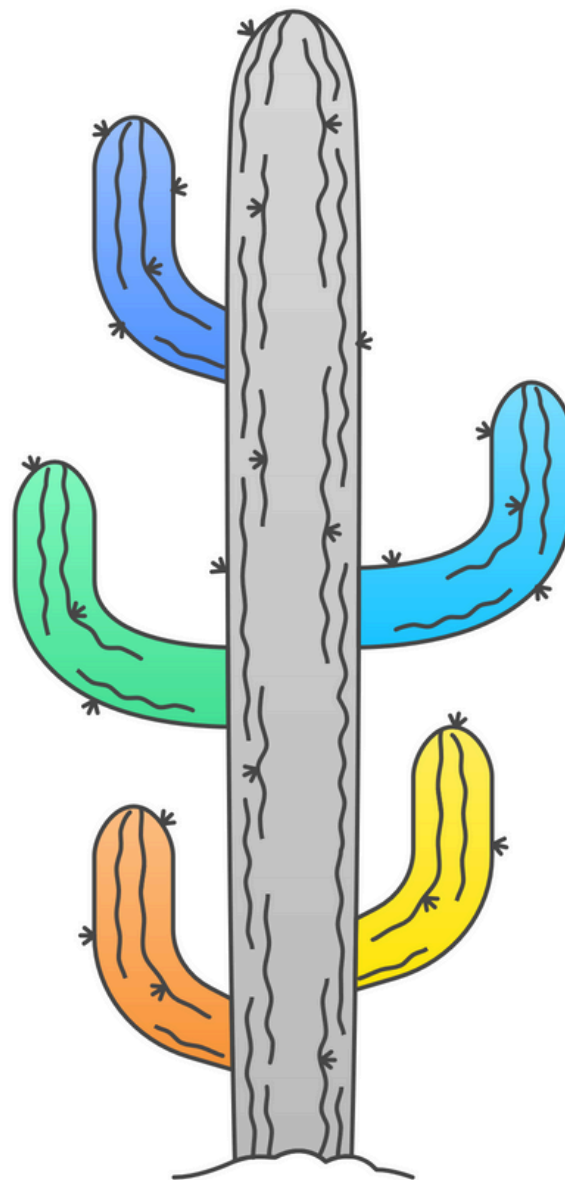
Push/pull synchronization is abstract

Checkout Process

Switching snapshots is confusing

Commit Storage

Linked commits are not clear



Merging Complexity

Internal merging process is unclear

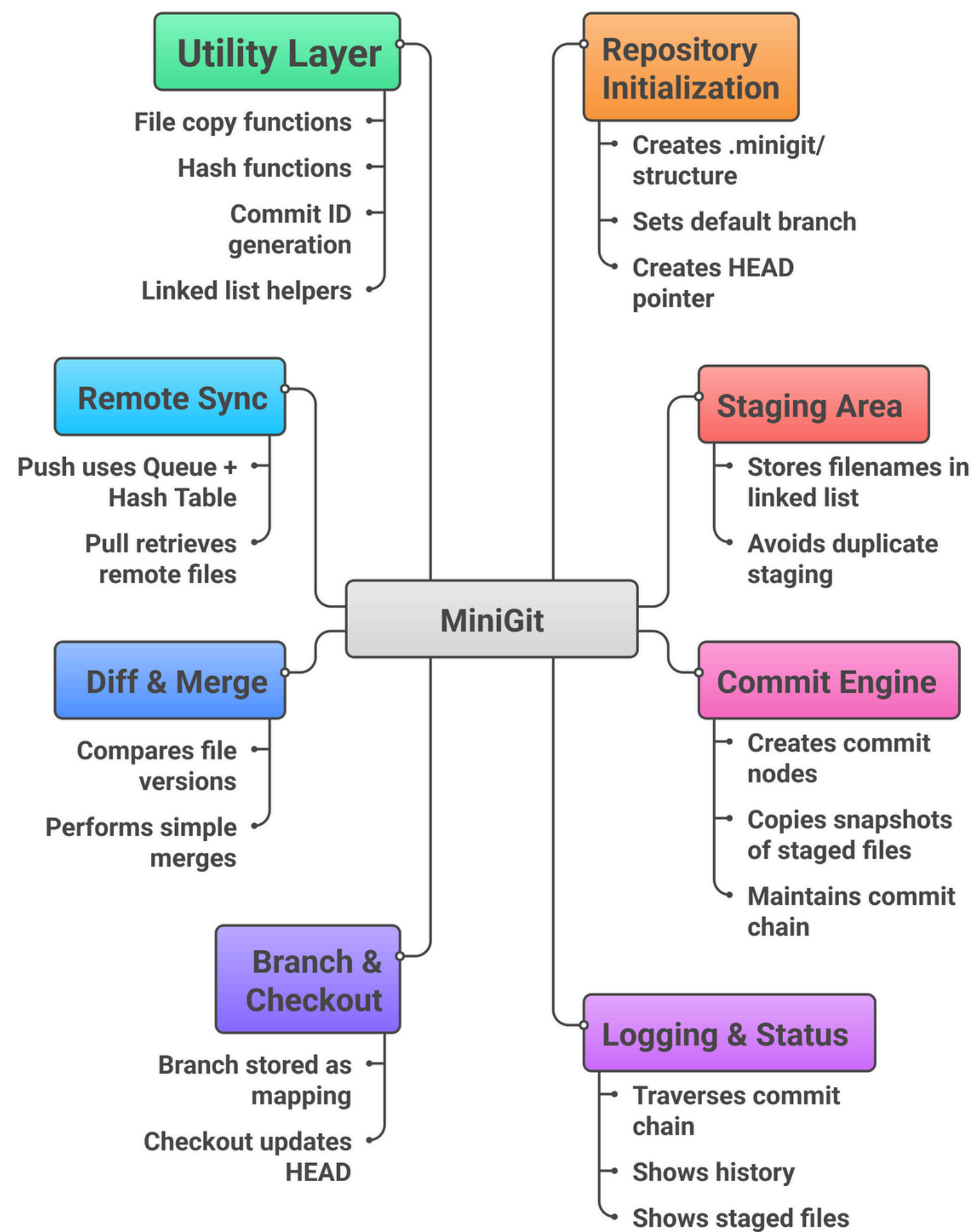
Branching Concepts

Branches point to commits vaguely

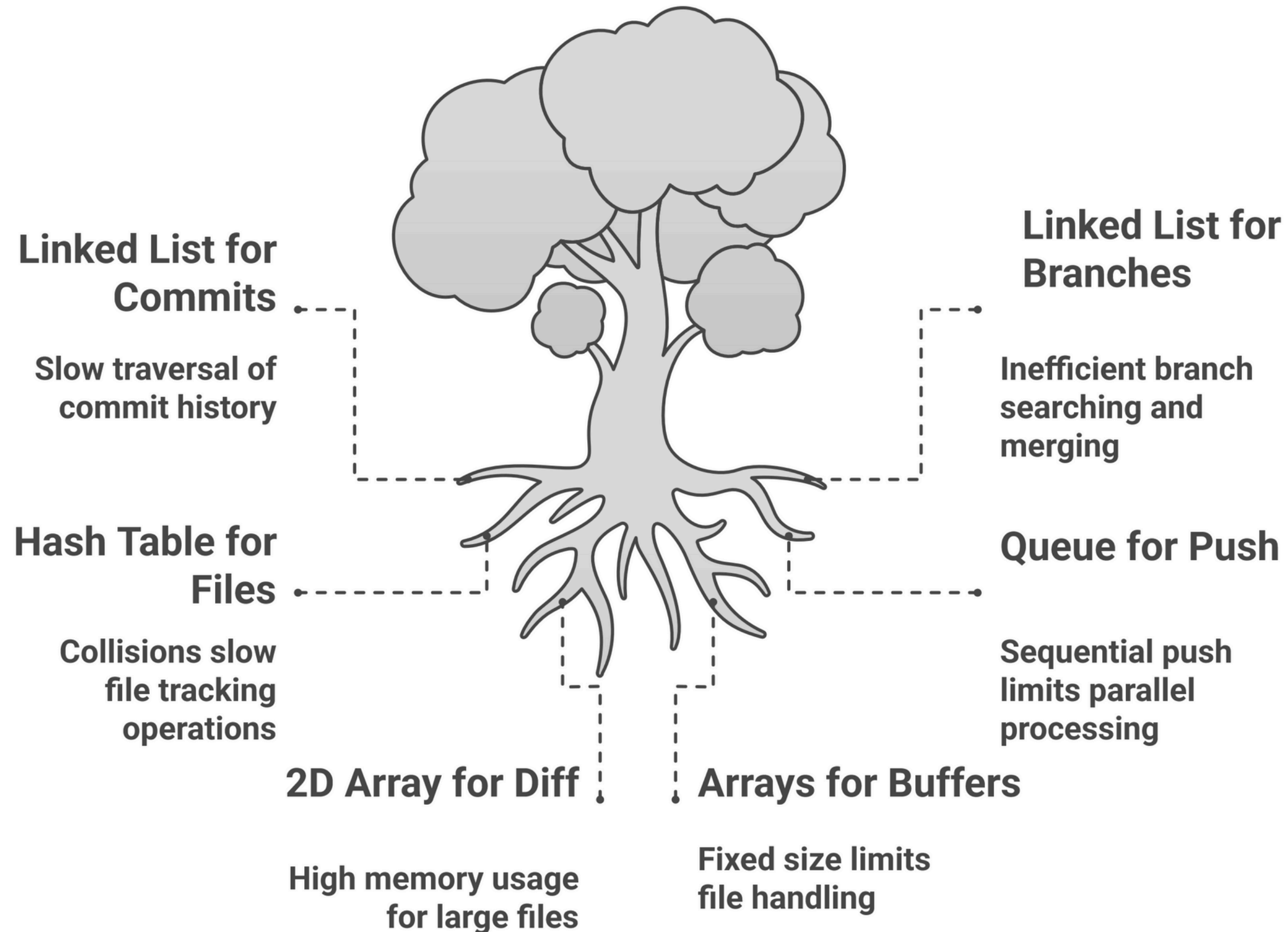
OUR SOLUTION

OBJECTIVES

- Build MiniGit, a simple version control system in C.
- Apply DSA concepts using linked lists, queues, and hash tables.
- Implement core Git-like features: init, add, commit, branch, merge, push, pull.
- Help students understand how version control works internally.
- Create a minimal, educational system for file tracking and version management.

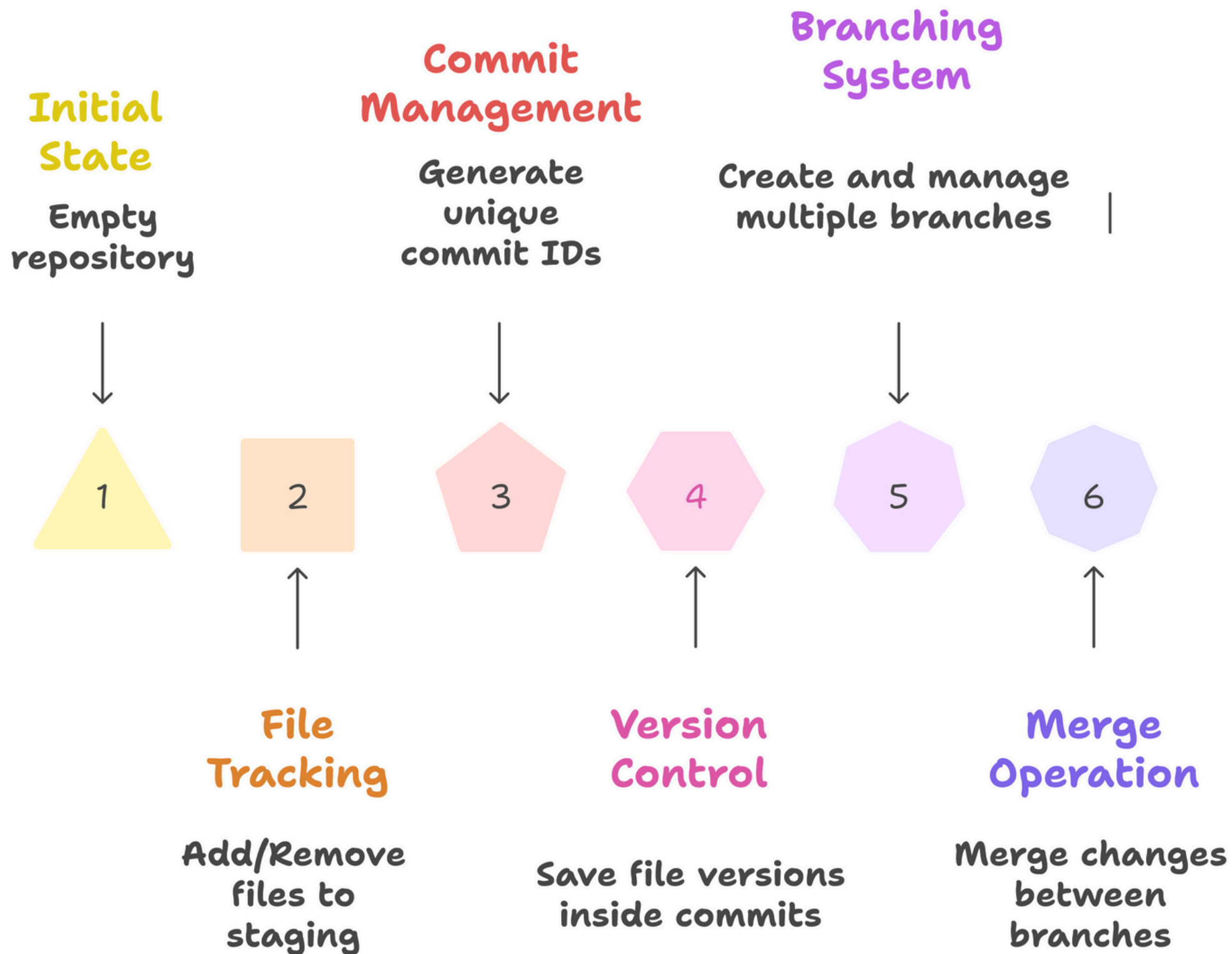


DATA STRUCTURES USED

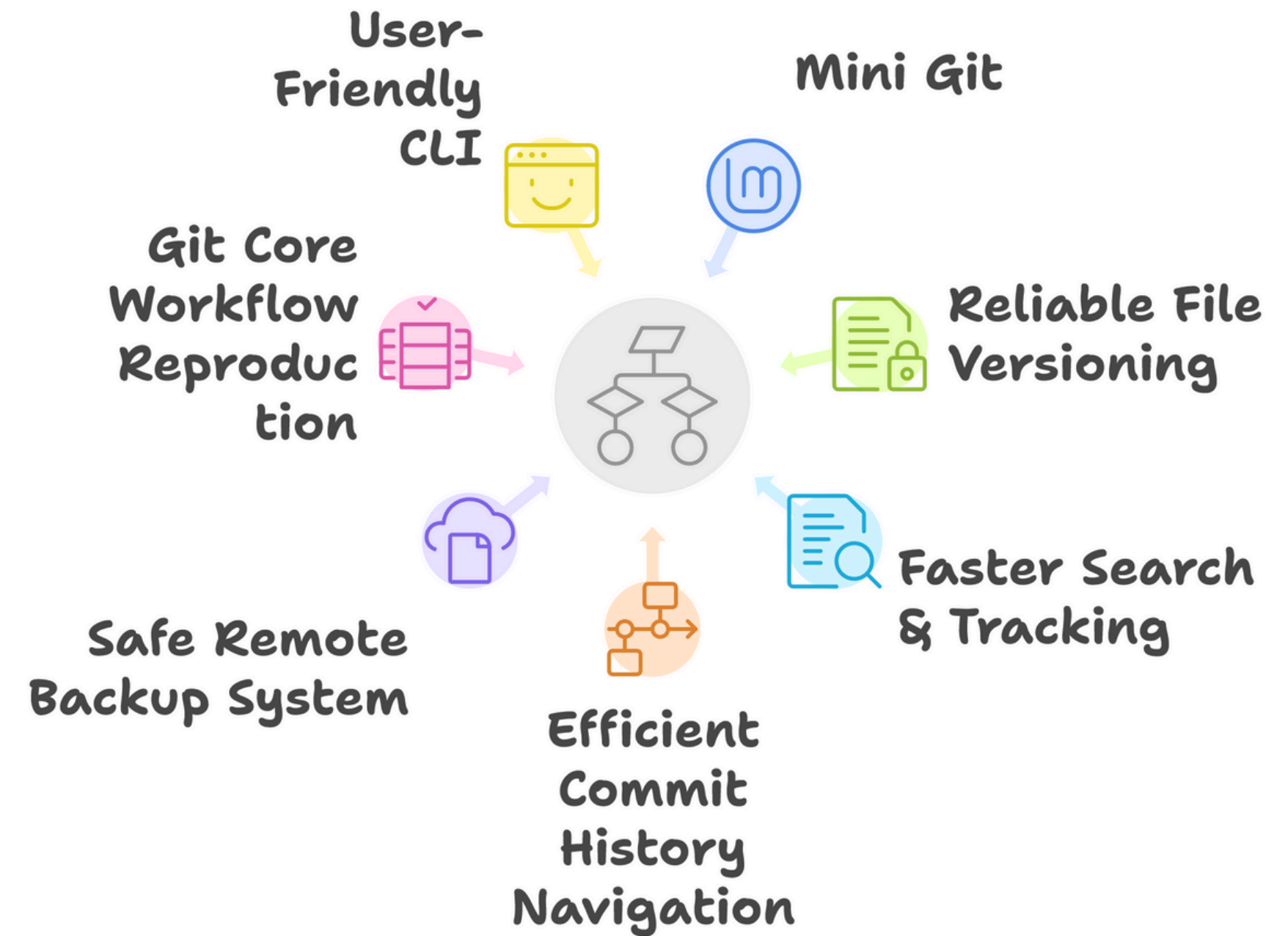


KEY FEATURES

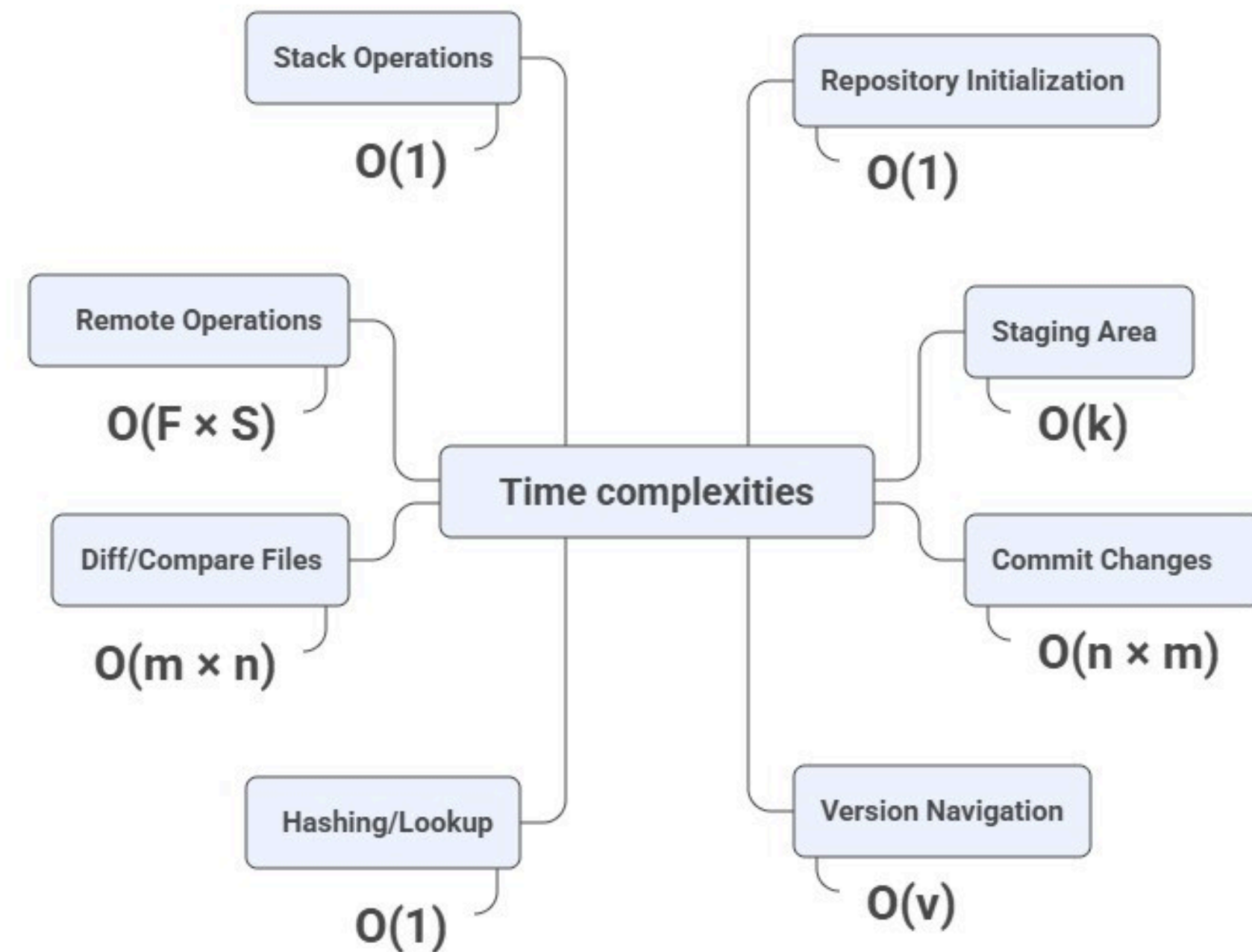
MiniGit Version Control System



OUTCOMES OF MINIGIT



Time complexity helps us understand how fast our MiniGit operations run as data grows. By analyzing complexities, we ensure our version control functions—commit, diff, checkout, push/pull—remain efficient even with large files or many versions. It also validates that the data structures we used (lists, hash tables, stacks, queues, DP tables) are the best fit for building a reliable, scalable MiniGit system



FUTURE SCOPE

In the future, MiniGit can be improved by adding real online push and pull using basic networking, making branching and merging work more like actual Git, and creating a simple GUI to visualize commits and branches. We can also optimize the diff algorithm for better performance and include basic authentication to make remote access more secure. These enhancements will make MiniGit more powerful, user-friendly, and closer to a real version control system

Studio Shodwe

THANK YOU

By Rufus Stewart

www.reallygreatsite.com